

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQ TEST (25 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Precision (BL3)	Assessment:	Assessment Tool 3 – MCQ TEST
Weightage:	10% of CIA	Total Marks:	25 (25 Questions × 1 Marks)
CO1 Statement:	Apply appropriate visualization methods to represent distributions, proportions, and relationships in data (BL3).		
Student Name:	V.Surya <u>Satvika</u>	Reg. No.:	192524254
Section / Batch:	B.Tech AI&DS	Date:	30-7-2026

Code of Conduct

I v.surya satvika(192524254) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: suryasatvika.v

Instructions

- This test contains 25 MCQ questions, each carrying 1 mark. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Distributions	Analyzing spread, skewness, density, and outliers	1–5	5
x–y Relationships	Exploring relationships and clustering using scatter-based methods	6–10	5
Correlation & <u>Heatmaps</u>	Identifying correlations and density patterns	11–15	5
Temporal Data Representations	Visualizing trends, seasonality, and time-based patterns	16–20	5
Geospatial & <u>Spatio-Temporal</u> Data	Mapping spatial data and combining space with time	21–25	5
GRAND TOTAL:			25 Marks

Annexure A – Multiple Choice Questions (MCQ Test)

1. Which visualization is commonly used to show parts of a whole?

- a) Line graph
- b) Pie chart
- c) Scatter plot
- d) Heat map

2. A logistics company maps delivery routes across cities. Why is geospatial visualization preferred?

- a) It improves audio quality
- b) It represents spatial relationships between locations
- c) It removes routing information
- d) It reduces geographic accuracy

3. A dataset contains values: 11, 13, 15, 17, 19, 21, 23, 90. Calculate the median and identify the likely skewness.

- a) Median = 21, symmetric distribution
- b) Median = 18, right-skewed distribution
- c) Median = 23, uniform distribution
- d) Median = 15, left-skewed distribution

4. A histogram comparing two stores shows one store has a much wider spread of sales values. What analytical conclusion is correct?

- a) Spread does not affect performance evaluation
- b) Both stores have equal consistency
- c) Wider spread implies higher average sales automatically
- d) That store experiences greater variability in sales performance

5. A boxplot has the following values: $Q1 = 40$, Median = 55, $Q3 = 70$. Calculate the IQR and evaluate the central spread.

- a) IQR = 55, indicating uniform spread
- b) IQR = 15, indicating no variability
- c) IQR = 30, indicating moderate variability in the middle 50%
- d) IQR = 70, indicating symmetric distribution

6. A density plot of rainfall data becomes flatter and wider after climate changes over several years. Which analytical interpretation is strongest?

- a) Flat density curves indicate absence of observations
- b) Rainfall patterns have become more variable and less concentrated
- c) Rainfall variability has reduced completely
- d) The average rainfall must have decreased

7. A company compares yearly sales distributions. Year 1 $\rightarrow$ Mean = 500, SD = 40; Year 2 $\rightarrow$ Mean = 500, SD = 100. Which analytical conclusion is strongest?

- a) Mean values alone determine spread
- b) Both years have identical distributions
- c) Year 1 has greater variability
- d) Year 2 sales fluctuate more widely around the average

8. Two distributions are compared using boxplots: Distribution A $\rightarrow$ Median = 55, Range = 20; Distribution B $\rightarrow$ Median = 70, Range = 50. Which analytical evaluation is most appropriate?

- a) Distribution B has higher central tendency but also greater spread
- b) Distribution A has more variability because its range is smaller
- c) Both distributions have equal consistency
- d) Range cannot indicate spread

9. Two boxplots are compared: Product A $\rightarrow$ Median lifetime = 8 years, IQR = 1 year; Product B $\rightarrow$ Median lifetime = 8 years, IQR = 4 years. Which evaluation is most valid?

- a) Equal medians imply equal variability
- b) IQR does not affect reliability interpretation
- c) Product A has more reliable lifetime consistency
- d) Product B is more stable because its IQR is larger

10. A dataset contains values: 22, 24, 25, 28, 30, 32, 35, 100. Calculate the median and evaluate the effect of 100 on the distribution.

- a) Median = 32, and distribution remains symmetric
- b) Median = 25, and 100 creates left skewness
- c) Median = 29, and 100 creates right skewness
- d) Median = 35, and no skewness exists

11. A histogram for monthly profits shows: 0–10k $\rightarrow$ 5 months, 10k–20k $\rightarrow$ 8 months, 20k–30k $\rightarrow$ 15 months, 30k–40k $\rightarrow$ 20 months, 40k–50k $\rightarrow$ 4 months. What analytical conclusion is most suitable?

- a) The distribution is strongly left-skewed
- b) Profits are concentrated mainly between 30k–40k
- c) No concentration pattern exists
- d) Profits are uniformly distributed

12. Two violin plots represent employee salaries: Department X $\rightarrow$ Mean = 50k, SD = 5k; Department Y $\rightarrow$ Mean = 50k, SD = 20k. Which evaluation is strongest?

- a) Department X has more spread because deviation is smaller
- b) Violin plots ignore standard deviation
- c) Department Y has greater salary variability
- d) Both departments have identical distributions

13. A dataset contains the values: 8, 10, 12, 14, 16, 18, 20, 60. Calculate the median and evaluate whether the distribution is skewed.

- a) Median = 18, distribution is symmetric
- b) Median = 15, distribution is right-skewed
- c) Median = 20, distribution is uniform
- d) Median = 14, distribution is left-skewed

14. A company visualizes internet traffic between servers using directed graphs. Why are arrows used on edges?

- a) To indicate data direction flow
- b) To improve brightness
- c) To reduce connectivity
- d) To hide nodes

15. A researcher compares rainfall patterns over ten years. Which temporal visualization supports trend analysis best?

- a) Line chart with time axis
- b) Pie chart
- c) Word cloud
- d) Static matrix

16. Assertion (A): Temporal visualizations can reveal seasonality patterns.

Reason (R): Data values are organized according to time progression.

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true, but R is false
- d) A is false, but R is true

17. A navigation application highlights roads with heavy traffic in red. What geospatial principle is applied here?

- a) Spatial intensity encoding
- b) Random clustering
- c) Audio normalization
- d) Binary formatting

18. In a network visualization of airline routes, hubs have many incoming and outgoing edges. What does this indicate?

- a) Low connectivity
- b) High network importance
- c) Data corruption
- d) Reduced traffic

19. A company compares two delivery systems: System A $\rightarrow$ Mean delivery time = 25 min, SD = 3; System B $\rightarrow$ Mean delivery time = 25 min, SD = 10. Which analytical conclusion is most valid?

- a) Both systems have identical variability
- b) Standard deviation cannot measure consistency
- c) System A is more consistent in delivery performance
- d) System B is more stable because it has higher deviation

20. A histogram shows frequencies of product defects: 0–2 defects $\rightarrow$ 60 machines, 2–4 defects $\rightarrow$ 25 machines, 4–6 defects $\rightarrow$ 10 machines, 6–8 defects $\rightarrow$ 5 machines. What analytical conclusion should be drawn?

- a) Defects are uniformly distributed
- b) Most machines produce high defects
- c) The histogram is bimodal
- d) Most machines produce very few defects, indicating right-skewness

21. A hospital analyzes patient waiting times using a histogram and notices that most values are concentrated in lower intervals while a few extend far to the right. What analytical interpretation can be made from this distribution?

- a) Waiting times follow a right-skewed distribution with a few extreme delays
- b) All patients experience identical waiting times
- c) The data follows perfect symmetry
- d) No variability exists in the dataset

22. A financial analyst uses a violin plot to compare monthly investment returns of two portfolios. One violin plot shows two clear density peaks. What does this most likely indicate about the portfolio returns?

- a) The returns contain two dominant groups or patterns
- b) The portfolio has zero variability
- c) The returns are uniformly distributed
- d) The dataset contains only outliers

23. Why do analysts use scatter plots for x–y data?

- a) To identify trends and relationships
- b) To display percentages only
- c) To organize files
- d) To remove variables

24. How does visualizing amounts help decision-making?

- a) By making quantity comparisons easier
- b) By reducing information
- c) By hiding differences
- d) By avoiding analysis

25. Why are distributions important in data analysis?

- a) They help understand how values are spread
- b) They remove patterns
- c) They reduce accuracy
- d) They avoid comparisons

Rubrics for Calculation

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks	
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding ; several incorrect responses	Lacks understanding; majority of answers incorrect		
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers		
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts		
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time		
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____